

JFXMaze Two-Page Summary

Programming II Final Project

Field	Value
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Project Title	JFXMaze
Displayed Game Name	NEON MAZE
GitHub Repository	https://github.com/miaralbarghouth/JFXMaze_neonmaze

Project Description

JFXMaze is a JavaFX desktop maze game. The user logs in, starts a randomly generated neon-style maze, collects points, uses optional hints, and tries to reach the exit. The project was built with Java 17, JavaFX 21, and Maven.

Required Features

- Basic application functions: playable maze, scoring, timer, pause menu, settings, and leaderboard.
- Authentication: user registration, login, logout, account editing, and account deletion.
- File processing: local account and leaderboard data are saved and loaded from disk.
- Maven configuration: dependencies and JavaFX plugin are defined in pom.xml.

Technical Highlights

- DFS-based random maze generation.
- BFS-based hint path calculation.
- Salted SHA-256 password hashing.
- JavaFX Canvas rendering for the main game screen.
- Two-language interface: English default and Turkish selectable in settings.

Installation

- 1 Clone the GitHub repository: https://github.com/miaralbarghouth/JFXMaze_neonmaze.git
- 2 Open the project folder.
- 3 Run `mvn javafx:run`.
- 4 Log in with the default test account `mk / mk` or create a new account.

Presentation Notes

- Show the login and registration flow.
- Start a game and demonstrate movement, scoring, pause, and hint.
- Open settings and show the language options.
- Explain how local file processing saves users and scores.
- Briefly explain DFS for maze generation and BFS for hints.

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